Numerical Analysis
Preparation for the Test – Supplementary Material

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1 Descriptive Statistics and Quartiles

Data Set

$$3,\ 5,\ 7,\ 8,\ 9,\ 10,\ 11,\ 13,\ 15,\ 16,\ 18,\ 20$$

Results

$$\text{Median} = \frac{10+11}{2} = 10.5,$$
$$Q_1 = \frac{7+8}{2} = 7.5, \quad Q_3 = \frac{15+16}{2} = 15.5.$$

2 Probability

Group	Low	High	Total
A	3	3	6
B	3	3	6

$$P(A \cap \text{Low}) = \frac{3}{12} = 0.25,$$
$$P(A) = \frac{6}{12} = 0.5,$$
$$P(\text{High} \mid A) = \frac{3}{6} = 0.5.$$

3 Hypothesis Testing

Data

$$78, 72, 69, 81, 75, 74, 68, 79, 83, 71, 77, 73$$

Hypotheses

$$H_0 : \mu = 70,$$
$$H_1 : \mu > 70.$$

Results

$$\bar{x} = 75, \quad s = 4.71, \quad t = 3.68,$$
$$p \approx 0.0018 \Rightarrow \text{Reject } H_0.$$

4 Type I and Type II Errors

Cereal boxes should average 500 g.

$$H_0 : \mu = 500,$$
$$H_1 : \mu < 500.$$

Type I: Conclude underfilling when correct.
Type II: Miss real underfilling.

5 Confidence Intervals

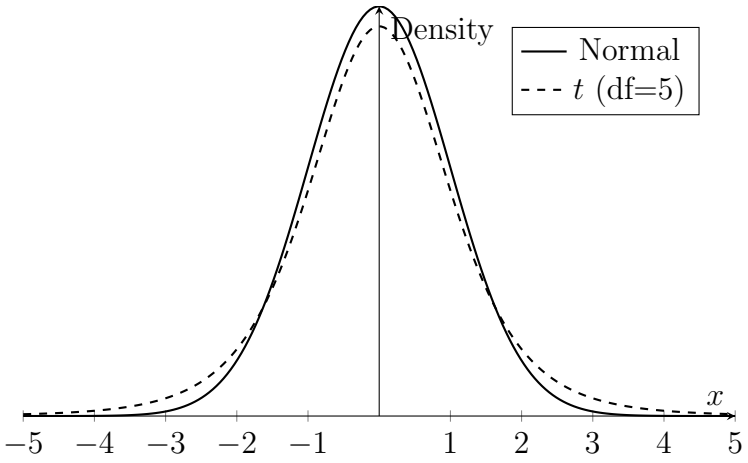
Data

$$4.2, 5.1, 3.8, 4.5, 4.9, 5.3, 4.7, 3.9, 4.6, 5.0, 4.3, 4.8$$

Results

$$\bar{x} = 4.425,\ s = 0.501,\ SE = 0.145,$$
$$\text{95\% CI (t)} : (4.11,\ 4.74),$$
$$\text{95\% CI (z, } \sigma = 0.5) : (4.14,\ 4.71).$$

6 Normal vs t Distribution



7 Vectors, Cross Product, and Matrices

Vectors

$$\mathbf{v} = \begin{bmatrix} 18 \\ 22 \\ 25 \\ 20 \\ 15 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} 4 \\ 6 \\ 8 \\ 5 \\ 3 \end{bmatrix}$$

$$2\mathbf{v} = \begin{bmatrix} 36 \\ 44 \\ 50 \\ 40 \\ 30 \end{bmatrix},$$

$$\mathbf{v} + \mathbf{u} = \begin{bmatrix} 22 \\ 28 \\ 33 \\ 25 \\ 18 \end{bmatrix},$$

$$\mathbf{v} \cdot \mathbf{u} = 549.$$

Cross Product

$$\mathbf{a} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \ \mathbf{b} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}, \ \mathbf{a} \times \mathbf{b} = \begin{bmatrix} -10 \\ 7 \\ 9 \end{bmatrix}.$$

Matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \ B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}, \ 3A = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}, \ \det(A) = -2.$$

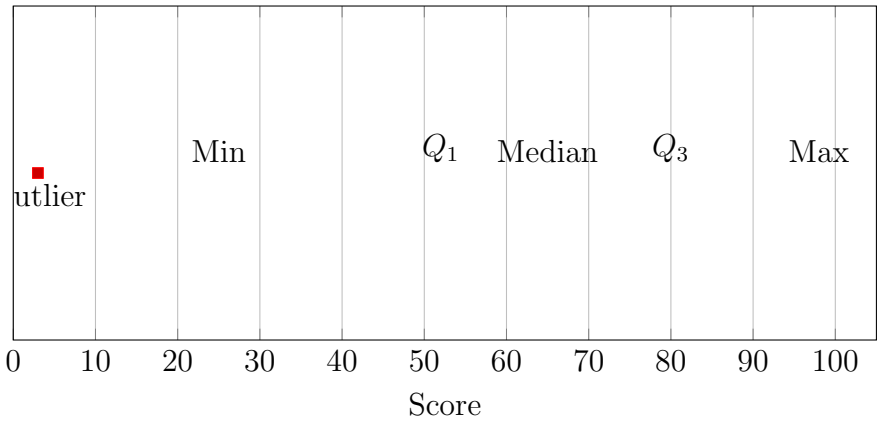
8 Box-and-Whisker Plot with Outlier

Data

$3, 25, 42, 45, 47, 48, 50, 52, 55, 56, 57, 58, 60, 62, 64, 65, 66, 68, 70, 72, 74, 75, 78, 80, 82, 85, 88, 90, 92, 95, 98$

Five-number Summary (non-outlier)

Min	25
Q_1	52
Median	65
Q_3	80
Max	98



9 Extra Practice Problems

A. Descriptive Statistics

Given

$$12, 15, 18, 21, 25, 27, 30, 32, 35, 38, 40$$

Find median, Q_1, Q_3, five-number summary, IQR, and outliers.

B. Probability

	Passed	Failed	Total
Male	32	18	50
Female	40	10	50

Compute marginal, joint, and conditional probabilities and test independence.

C. Hypothesis Testing

Claim: $\mu = 200$. Data:

$$192, 198, 201, 205, 190, 185, 210, 199, 203, 197, 194, 196, 202, 204, 193.$$

Test at $\alpha = 0.05$ and explain Type I/II errors.

D. Confidence Interval

Given $\bar{x} = 1.98, s = 0.12, n = 20$. Construct 95% CI and interpret.

E. Vectors

$$\mathbf{a} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \ \mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix}, \ k = -2.$$

Compute $k\mathbf{a}$, $\mathbf{a} + \mathbf{b}$, $\mathbf{a} \cdot \mathbf{b}$, $|\mathbf{a}|$, and angle.

F. Cross Product

Using vectors in E, compute $\mathbf{a} \times \mathbf{b}$ and its magnitude.

G. Matrix Operations

$$M = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}, \ N = \begin{bmatrix} 1 & 4 \\ -2 & 5 \end{bmatrix}.$$

Compute $M + N$, $2M$, MN, $\det(M)$ and invertibility.

H. Boxplot

Data:

$$24, 28, 30, 32, 35, 36, 38, 40, 42, 45, 110$$

Find five-number summary, outliers, skewness, and effect on mean/median.